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Dear Editor-in-Chief Professor May Yuan:

We are pleased to submit our manuscript entitled “Measuring causal strengths from spatial cross-sectional data by geographical cross mapping cardinality".

This study proposed a novel Geographical Cross Mapping Cardinality (GCMC) model developed to measure causal strengths from spatial cross-sectional data, addressing a critical gap in Earth system science, where there is a lack of data-driven models capable of examining causation from a spatial perspective.

The developed GCMC model provides a reliable methodological framework for measuring causal strengths and identifying the causal link, with broad applicability across various fields. Following the construction of benchmark systems to evaluate the effectiveness of GCMC, we further selected three real-world case studies to demonstrate its practical applicability and examine its robustness to observational noise. We believe that our proposed new model can provide a significant contribution to the field of GIS and spatial analysis. We look forward to the opportunity to contribute to this journal.

Author contributions: Wenbo Lv, Shaoqing Dai, and Yongze Song conceived the idea and designed the study; Wenbo Lv and Wufan Zhao collaborated on the implementation of the GCMC model in this manuscript; Wen Yi, Yumiao Xiao, and Nan Jia collected and analyzed the data; Wenbo Lv and Yongze Song wrote the first draft, and all authors contributed to the writing and editing of the final manuscript.

The authors declare no competing interests.

Please do not hesitate to contact me should you require additional information. We appreciate your kind consideration.

Sincerely yours,
Yongze

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